



State of health estimation of lithium-ion batteries based on multi-health features extraction and improved long short-term memory neural network

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ABSTRACT

Accurate state of health estimation of lithium-ion batteries is essential to enhance the reliability and safety of a battery system. However, the estimation accuracy based on a data-driven model is degraded by one health feature and incorrect hyper-parameters selection. This paper develops a battery state of health estimation method based on multi-health features extraction and an improved long short-term memory neural network. To accurately describe the aging mechanism of batteries, health features are extracted from battery data, such as time features, energy features, and incremental capacity features. The correlation between multi-health features and state of health is evaluated by the grey relational analysis. Aiming at the problem that the hyper-parameters of a neural network model are difficult to select, an improved quantum particle swarm optimization algorithm is developed to correctly obtain the hyper-parameters. The experimental results show that the mean absolute error, mean absolute percentage error, and root mean square error of this method are all within 1%, which is much lower than other methods, with high state of health estimation accuracy and robustness.

1. Introduction

Lithium-ion (Li-ion) batteries have been gradually popularized in the field of energy storage and electric vehicles due to their advantages, such as high energy density, long cycle life, and low self-discharge rate [1–4]. However, Li-ion batteries will age and degrade after multiple charging and discharging cycles, which can lead to problems such as liquid leakage and partial short circuits and eventually explosion and fire [5,6]. Therefore, it is necessary to accurately estimate the State of Health (SOH) and replace the aging battery in time. However, it is a challenge to accurately estimate the SOH of a Li-ion battery due to its complex and coupled internal aging mechanism [7,8].

At present, SOH estimation methods are mainly divided into three categories: direct measurement methods, model-based methods, and data-driven methods [9]. The direct measurement methods, including the Coulomb counting method, ampere-hour integral method, and impedance spectroscopy method, are to measure battery parameters directly related to battery aging in a laboratory environment, such as battery capacity and internal resistance. The aging characteristics of Li-ion batteries are studied by electrochemical impedance spectroscopy

technology [10]. This method is easy to use in offline measurement conditions but difficult to apply in online applications. Model-based methods mainly include the electrochemical model and equivalent circuit model (ECM). A method for identifying and selecting parameters of a pseudo-2-dimensional (P2D) model is proposed [11], which changes significantly with battery aging. Electrochemical model can be used to estimate the SOH due to its merits, such as clear physical meaning. However, it is not suitable for online estimation of SOH because of its complex model and the complicated model parameter identification. In ECM, some nonlinear circuit elements are used to simulate the internal working state of the battery. And combined with the filter algorithms, it is used for battery parameter identification and SOH estimation. For example, the ECM combined with a dual extended Kalman filter is used to estimate SOH of a battery [12]. Compared with the electrochemical model, the complexity of the ECM is significantly reduced. The SOH estimation accuracy based on the ECM is greatly affected by the parameter identification and filtering algorithms, which reduces its robustness [13].

The data-driven methods can estimate SOH of the batteries with strong flexibility and expansibility due to not considering the internal

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mechanism and the complex models of the batteries [14,15]. In these methods, only the battery aging health features (HFs) should be extracted from the battery status data, such as voltage, current, and temperature, and machine learning or deep learning algorithms are used to fit the nonlinear mapping relationship between HFs and SOH to achieve SOH estimation. Common data-driven methods include but are not limited to neural network (NN) [16,17], support vector machine (SVM) [18,19], Gaussian process regression (GPR) [20,21], autoregressive model [22,23], etc. For example, a two-layer structured SVM with radial basis functions is used to estimate SOH [24]. A battery degradation model based on a dual GPR framework is proposed to estimate the battery SOH and remaining useful life (RUL), respectively [25]. Moreover, as one of data-driven methods, the NNs as a data-based nonlinear method are suitable for complex nonlinear modeling problems with multiple correlated features. An improved particle swarm optimization algorithm is used to optimize the weights and thresholds of back-propagation neural networks (BPNNs), which can achieve precise estimation of SOH [26]. An improved grey wolf optimization algorithm is presented to optimize the initial parameters of radial basis neural networks (RBFNNs), which can illustrate both the overall trend and local fluctuations of SOH [27].

Compared with the feed-forward NN such as BPNNs, the recurrent neural networks (RNNs) can retain some important information of the network and update this information in a certain time by the cyclic feedback mechanism of network parameters [28]. However, the SOH estimation accuracy using the RNNs is degraded by its disadvantages such as gradient disappearance or even explosion. As one of the improved RNNs, the long short-term memory (LSTM) NN can overcome the above problem and retain long and short sequence dependencies. As an effective method to deal with time series data such as battery degradation, the LSTM has great application prospects in SOH estimation [29,30]. For example, the LSTM combined with transfer learning is presented to predict the SOH of different types of batteries [29]. A hybrid model based on the attention mechanism and bidirectional LSTM is proposed to accurately estimate the SOH [30].

For accurate estimation of SOH, in addition to improving the accuracy and efficiency of the SOH estimation algorithm, it is also very important to accurately extract HFs that can characterize the battery aging. According to the characteristics of shorter and shorter discharge time in the repeated charge and discharge cycle of the batteries, the discharge voltage difference at equal time intervals is selected as one HF to estimate SOH [31]. The battery charging capacity during the battery voltage changes from 3.6 V to 4.2 V during the constant current (CC) charging stage; this capacity is extracted as one HF for the SOH estimation [32]. Three features—namely the voltage difference and time interval during the CC stage and the current difference during the constant voltage (CV) stage—are extracted in the charging process [33]. The energy of the CV charging stage and the energy of the equal discharge voltage difference are extracted to estimate the battery SOH [34]. The peak position and peak valley position characteristic parameters of the incremental capacity (IC) curve are extracted as HFs from the static charging curve of the batteries [35].

However, the SOH estimation accuracy using the data-driven methods is degraded due to the shortcomings in terms of HF extraction and models. In terms of HF extraction, only a single feature or the same type of feature is extracted as one HF in many studies. It is difficult to accurately and comprehensively reflect the complex degradation trend of Li-ion batteries using one HF. In terms of the models, the generalization performance of the models should be poor when the internal structures and the parameters in the data-driven models cannot flexibly be set according to the batteries' feature data. For example, hyper-parameters such as learning rate and the number of neurons in the hidden layer of the LSTM model are usually set based on human experience, which degrades the accuracy and efficiency of the SOH estimation.

To overcome the above problems, a SOH estimation method of Li-ion

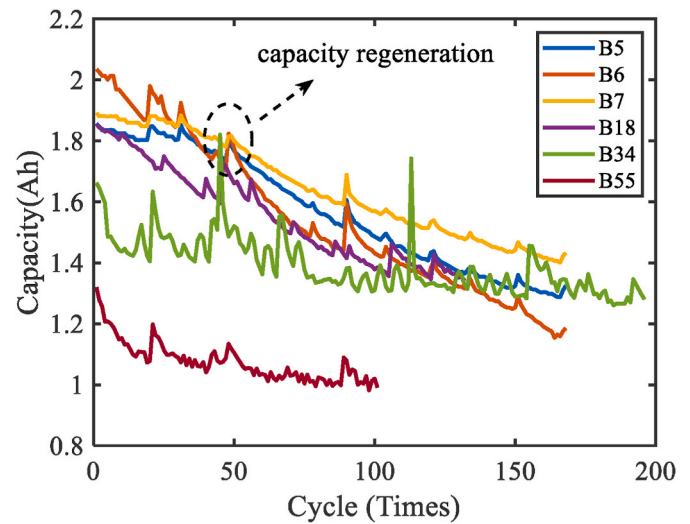


Fig. 1. Capacity degradation of the batteries.

batteries based on multi-HF extraction and an improved LSTM model are developed. The main innovations of the developed method in this paper include: (1) in order to accurately describe the battery ageing process, ten HFs are extracted from the battery data and divided into three categories, including time features, energy features, and IC features; (2) to determine the optimal hyper-parameters of LSTM, an improved quantum particle swarm optimization (IQPSO) algorithm is presented to further improve the global searching ability of the standard QPSO; (3) an IQPSO-LSTM model with the extracted ten HFs is established to accurately estimate the SOH of the Li-ion batteries.

The rest of this paper is organized as follows: Section 2 introduces the experimental dataset and the extraction process of health features, Section 3 presents the SOH estimation model based on IQPSO-LSTM, Section 4 shows the accuracy and effectiveness of the developed SOH estimation method in comparison with other methods, and Section 5 discusses the conclusions.

2. Battery data analysis and multi-health features extraction

2.1. The definition of SOH

For Li-ion rechargeable batteries, the maximum available capacity will decrease and internal resistance will increase when the number of charge and discharge cycles increases. Therefore, the SOH of Li-ion batteries is often defined in terms of the capacity or the internal resistance [36]. To quantitatively illustrate the battery's decay process, the SOH definition of Li-ion batteries is described in terms of a battery's capacity in this paper as follows:

$$SOH = \frac{C_{current}}{C_{rated}} \times 100\% \quad (1)$$

where $C_{current}$ and C_{rated} are the current capacity and rated capacity of the batteries, respectively.

2.2. Description of experimental data

The Li-ion battery dataset from the National Aeronautics and Space Administration (NASA) Ames Prognostics Center of Excellence (PCoE) was used in this paper [37], and the batteries were commercial rechargeable 18,650 Li-ion batteries with nominal capacity of 2 A h. In this paper, B5, B6, B7, B18, B34, and B55 batteries are selected, the charge-discharge cycle experiments of B5, B6, B7, B18, and B34 were carried out at 24 °C, and B55 were carried out at 4 °C, where the charging process was divided into the CC charging mode and the CV

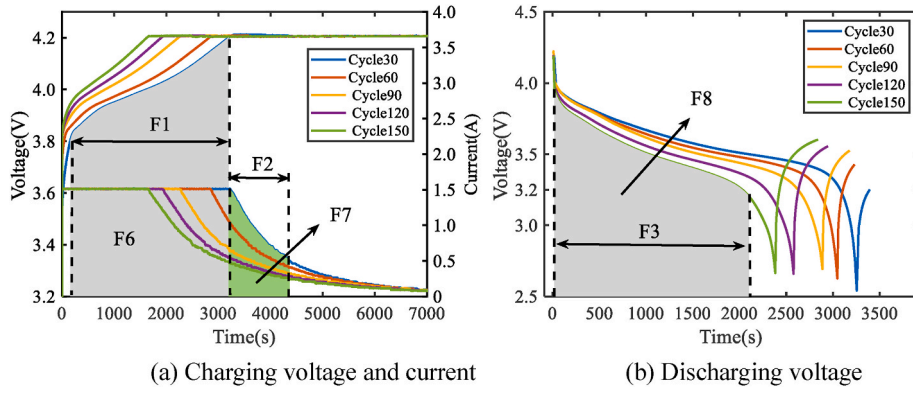


Fig. 2. Current and voltage variation vs. the time in charging and discharging process.

charging mode. First, the batteries were charged in the CC charge mode at 1.5 A until the battery voltage reached 4.2 V, then the batteries were charged in the CV charge mode until the battery current dropped to 20 mA. After a rest time of 2 h, the batteries were discharged at the constant current of 2 A until the battery voltage dropped to 2.7 V, 2.5 V, 2.2 V, 2.5 V, and 2.5 V for batteries B5, B6, B7, B18 and B55, respectively. But B34 was discharged at the constant current of 4 A until its voltage arrived to 2.2 V. The capacity degradation of these batteries is shown in Fig. 1.

2.3. Multi-health features extraction

As can be seen from Fig. 1, the battery capacity shows a nonlinear decrease with the increase of the number of cycles, and the battery capacity curve illustrates the characteristics of the fluctuations and local regeneration, especially for the B34 battery in 4 A discharge current. This is because when the lithium-ion battery is resting, some of the unstable compounds gathered near the electrode are re-decomposed into lithium ions, which can be combined with the electrode. Then, the available capacity of the battery in next cycle can be increased and results in the occurrence of capacity regeneration. The SOH estimation accuracy is greatly affected by this nonlinear relationship between the battery capacity and the cycles. Therefore, it is necessary to extract some robust and accurate HFs from the external parameters of the battery to effectively estimate the SOH of the batteries. To accurately describe the battery degradation variation, in this paper, ten HFs—which are listed F1~F10 and divided into time HFs, energy HFs, and incremental capacity (IC) HFs—are extracted from the battery data in the charging and discharging process, such as voltage, current, temperature, and test time.

2.3.1. The extraction of time HFs

As one of the monitoring parameters of the battery, the charging and discharging time can simply and intuitively reflect the aging of the battery. As shown in Fig. 2 (a), when the number of cycles increases in the charging process, the charging time decreases in the CC phase, but it increases in the CV phase. For example, the charging time in the CV phase is about 1500 s when the battery cycle is 30, but the charging time in the CV phase is about 3000 s while the battery cycle is 150. This phenomenon is caused by the decay of the lithium battery material over time. Moreover, since the voltage variation between 3.8 V and 4.2 V in the CC stage and the current variation between 1.5 A and 0.5 A in the CV stage are significant, the time interval of the equal voltage difference when the battery charging voltage changes between 3.8 V and 4.2 V and the time interval of the equal current difference when the battery charging current changes between 1.5 A and 0.5 A are selected as the time HFs, denoted as F1 and F2, respectively. Fig. 2 (b) shows the battery discharging voltage variation with the discharge time. It can be seen that the discharging time of the battery gradually decreases when the cycle number increases, so the discharge time can be selected as one of the HFs. In practical applications, the batteries usually operate at a voltage range from 4.0 V to 3.2 V in the discharging process to protect the batteries, and the discharging voltage variation is stable. In this paper, the time interval of the equal voltage difference when the battery discharging voltage changes between 4.0 V and 3.2 V is selected as one of the time HFs, denoted as F3.

In addition, as a major factor affecting the reliability and safety of Li-ion batteries, the battery temperature has a significant influence on the electrochemical reactions inside the battery. As shown in Fig. 3, the larger the number of cycles, the shorter the time for the battery temperature to reach its peak. In this paper, the times when the battery temperature reaches its peak during the charging process and the discharging process are extracted as the time HF, denoted as F4 and F5,

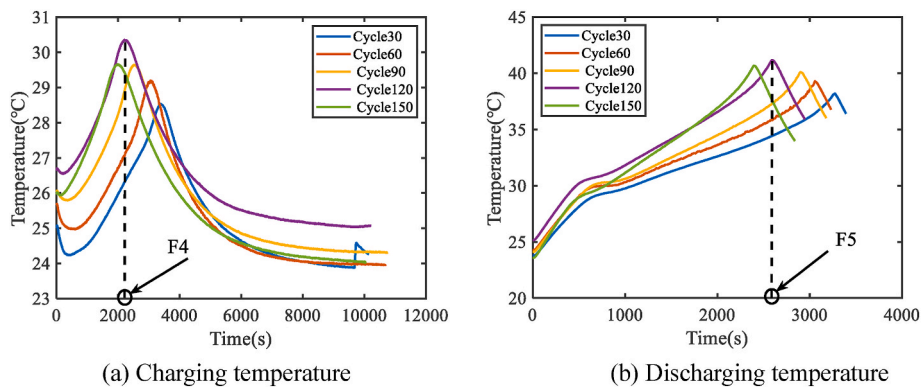


Fig. 3. The charging and discharging temperature vs. times.

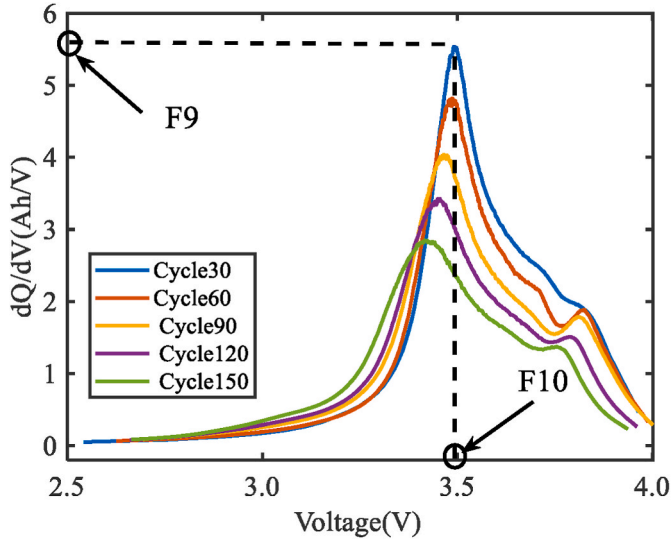


Fig. 4. Incremental capacity vs. battery voltage.

respectively.

2.3.2. The extraction of energy HF's

The charging and discharging process of the lithium batteries is a mutual energy conversion between electrical energy and chemical energy. As the number of charge/discharge cycles increases, the energy that can be charged and discharged in the battery will gradually decrease. As shown in Fig. 2 (a), the area consisting of the voltage curve in the CC stage and the horizontal and vertical coordinates decreases, and the area consisting of the current curve in the CV stage and the horizontal and vertical coordinates increases with the number of cycles,

respectively. It means that the charging energy decreases in the CC stage and the charging energy increases in the CV stage. In the discharging process, the area composed of the discharge voltage curve and the abscissa time is gradually decreasing, meaning that the discharge energy is decreasing. In this paper, three energy HF's are extracted to describe the decline of the batteries' energy: energy of equal charge voltage difference (EECV), energy of equal charge current difference (EECC), and energy of equal discharge voltage difference (EEDV), and denoted as F6, F7, and F8, respectively. The mathematical representations of the three HF's can be expressed as follows:

$$EECV = \int_{t_1}^{t_2} V_{CC}(t) \cdot I_{CC}(t) dt \quad (2)$$

where t_1 and t_2 are the times when the charging voltage reaches 3.8 V and 4.2 V, respectively; $V_{CC}(t)$ and $I_{CC}(t)$ represent the charging voltage and charging current in the voltage interval, respectively.

$$EECC = \int_{t_3}^{t_4} V_{CV}(t) \cdot I_{CV}(t) dt \quad (3)$$

where t_3 and t_4 are the times when the charging current reaches 1.5 A and 0.5 A, respectively; $V_{CV}(t)$ and $I_{CV}(t)$ represent the charging voltage and charging current in the current interval, respectively.

$$EEDV = \int_{t_5}^{t_6} V_{dis}(t) \cdot I_{dis}(t) dt \quad (4)$$

where t_5 and t_6 are the times when the discharge voltage reaches 4.0 V and 3.2 V, respectively; $V_{dis}(t)$ and $I_{dis}(t)$ represent the discharging voltage and discharging current in the voltage interval, respectively.

2.3.3. The extraction of incremental capacity (IC) HF's

In this paper, the IC HF's are extracted by the incremental capacity analysis method, which can convert the voltage into an IC curve with

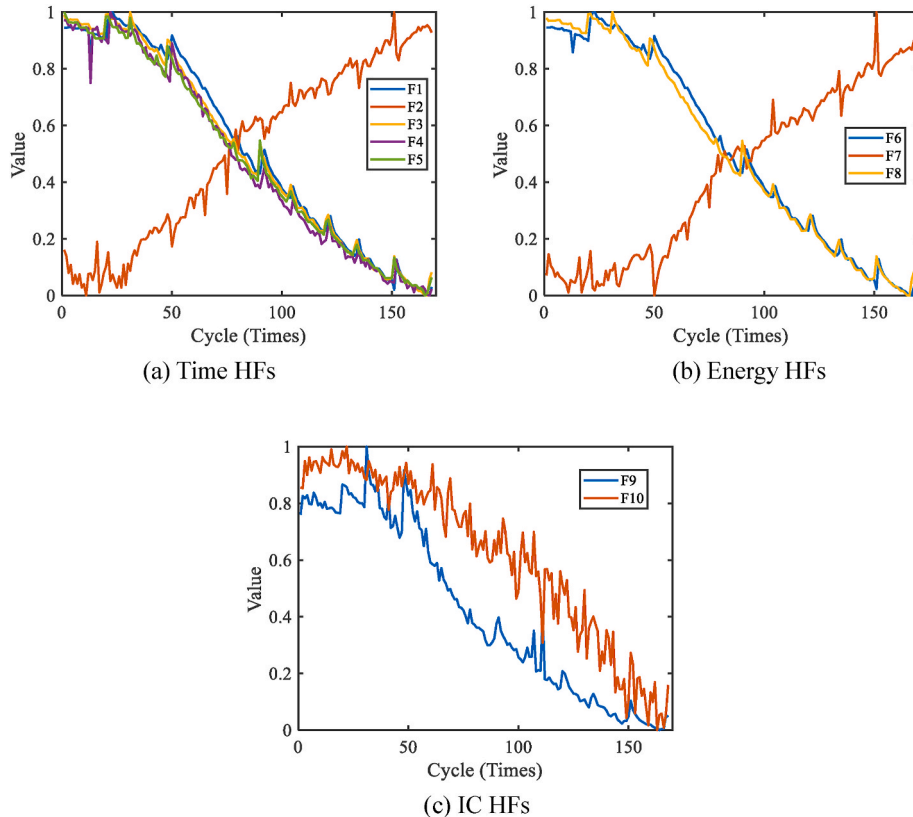


Fig. 5. The ten normalized HF's of B5.

Table 1

The correlation degree of the HF and the SOH.

	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10
B5	0.9238	0.8630	0.9789	0.9439	0.9984	0.9265	0.8677	0.9739	0.8977	0.9213
B6	0.6109	0.5398	0.8283	0.7668	0.9884	0.6177	0.5233	0.7994	0.6449	0.7086
B7	0.9245	0.8215	0.9730	0.9245	0.9927	0.8243	0.9281	0.9656	0.8573	0.8812
B18	0.9303	0.8715	0.9731	0.9516	0.9947	0.8725	0.9321	0.9668	0.8752	0.9120
B34	0.8137	0.8104	0.9883	0.8588	0.9390	0.8122	0.8191	0.9688	0.8062	0.8300
B55	0.9051	0.7737	0.9790	0.6451	0.9683	0.9063	0.9398	0.9807	0.8582	0.9404

obvious characteristics based on the battery voltage and current parameters. The IC curve can illustrate the phase transition characteristics of the battery during the intercalation process of the Li-ion active materials, including the Li-ion information participating in electrochemical reactions in different aging states of the battery [38]. In the CC discharge process of the Li-ion batteries, the IC calculation formula can be expressed as:

$$\frac{dQ}{dV} = \frac{dQ}{dT} \frac{dT}{dV} = I \cdot \frac{dT}{dV} \quad (5)$$

where Q , V , I , T are the discharging capacity, the discharging voltage, the discharging current, and the discharging time, respectively.

Fig. 4 shows the relation of the IC versus battery voltage. It can be found that the peak value of the IC gradually decreases and shifts to the left when the number of cycles increases. The reason for this phenomenon is that the ability of the internal active material embedding the lithium gradually decreases in the aging process. In this paper, the peak value and its corresponding positions of the IC curve are extracted as the IC HF, denoted as F9 and F10.

According to the features extraction method described in Section 2.3, ten HF are extracted from the NASA battery dataset and normalized to the interval of [0,1] (the normalization process is shown in Section 3.4). Taking the B5 battery as an example, the ten normalized HF are divided into three categories as shown in Fig. 5. Fig. 5(a) shows the extracted time HF, including 5 HF named F1~F5. The F1, F3, F4, and F5 decrease with the number of cycles, but the F2 increases with the number of the cycles due to the increasing time interval of the equal current difference. Fig. 5(b) shows the extracted energy HF, including three HF named F6~F8. The F7 for EECC gradually increases with the number of cycles, but the F6 and F8 gradually decrease with the number of the cycles. Fig. 5(c) shows the extracted IC features, including two HF named F9~F10 that decrease with the number of cycles.

2.4. Grey correlation analysis of the multi-HFs

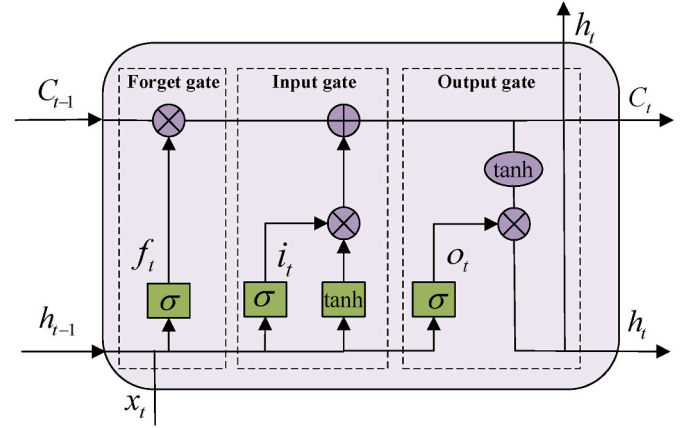
To quantitatively analyze the correlation degree of the multi-HFs and the SOH, a grey correlation analysis method [39] based on the grey system theory is used in this paper. Using this analysis method, the similarity and difference between the reference sequence and the comparison sequence can be quantitatively analyzed. The calculation of the grey correlation coefficient can be presented as follows:

$$\xi_i(k) = \frac{\min_{i,k} |y(k) - x_i(k)| + \rho \min_{i,k} |y(k) - x_i(k)|}{|y(k) - x_i(k)| + \rho \max_{i,k} |y(k) - x_i(k)|} \quad (6)$$

where $\xi_i(k)$ is the grey correlation coefficient between the i -th HF and the SOH, $k = 1, 2, \dots, n$, n represents the number of the battery cycles, $y(k)$ is the SOH sequence, $x_i(k)$ is the HF sequence, and ρ is the resolution coefficient and is set as 0.5. The grey correlation degree can be obtained by calculating the average value of the grey correlation coefficient as follows:

$$\gamma_i = \frac{1}{n} \sum_{k=1}^n \xi_i(k) \quad (7)$$

The range of γ_i is [0,1]. The closer the value of γ_i is to 1, the stronger

**Fig. 6.** The architecture of the LSTM.

the correlation between the reference sequence and the comparison sequence. According to the formulas in (6) and (7), the grey correlation degree between ten HF and the SOH is obtained, which is listed in Table 1. In general, when the correlation value is greater than 0.8, it indicates that the correlation between the HF and the SOH are strong. When the correlation degree is between 0.5 and 0.8, it indicates that the correlation between the HF and the SOH are moderate. Otherwise, when the correlation value is less than 0.5, it means that the correlation is weak. Table 1 indicates that the extracted HF have a strong correlation with the SOH because most of the correlation degrees are greater than 0.8, and the ten HF can be used as the inputs for the SOH estimation model.

3. Methodology

3.1. LSTM neural network

As an improved RNN, the LSTM has a unique memory cell mechanism that can remove or add some information from or into the cell state by the three gate units—forgetting gate, input gate, and output gate—to achieve the information protection and control [40]. The architecture of the LSTM is shown in Fig. 6.

The core step of the LSTM can be presented as follows.

- (1) Decide what information should be discarded from the unit state in the forgetting gate f_t :

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (8)$$

where σ is the sigmoid function, W_f and b_f represent the weight matrix and bias term of the

Forgetting gate, h_{t-1} is the output of the unit state at time $t-1$, and x_t is the input of the unit state at time t .

- (2) Decide what information should be stored in the internal state in two steps. First, update the input gate i_t .

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (9)$$

where W_i and b_i are the weight matrix and bias items of the input gate.

Secondly, achieve a candidate vector $\tilde{C}_t$,

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (10)$$

where W_c and b_c are the weight matrix and the bias item of $\tilde{C}_t$,

(3) Update the unit state

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (11)$$

where C_{t-1} is the unit state at $t-1$ time;

(4) Attain the output of LSTM

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (12)$$

$$h_t = o_t * \tanh(C_t) \quad (13)$$

where o_t is the Sigmoid layer of the output gate, h_t is the output of the LSTM at time t , and W_o and b_o are the weight matrix and bias item of the output layer, respectively.

3.2. Standard quantum particle swarm optimization (QPSO)

In the standard particle swarm optimization (PSO) algorithm, the particle randomness and its global convergence performance are usually degraded by the size and direction of its particle velocity determined by the velocity updating formula. Compared with the PSO, the QPSO shows randomness of the particle position and can avoid falling into the local optimum because it removes the movement direction of the particle and updates the particle position in spite of the previous motion of the particle [41]. The core iterative process of the QPSO can be expressed as follows.

(1) Initialization of the population

$$\begin{cases} x_i(t) = [x_{i1}(t), x_{i2}(t), \dots, x_{iD}(t)] \\ P_i(t) = [P_{i1}(t), P_{i2}(t), \dots, P_{iD}(t)] \\ P_g(t) = [P_{g1}(t), P_{g2}(t), \dots, P_{gD}(t)] \end{cases} \quad (14)$$

where the population is composed of M particles in D dimensional space, $i = 1, 2, \dots, M$, $g \in [1, 2, \dots, M]$, $x_i(t)$ is the position of the i -th particle at the t -th iteration, $P_i(t)$ is the historical optimal position of the i -th particle, and $P_g(t)$ is the optimal position of the population.

(2) Search Iteration Strategy

With the iterative evolution of the population, the particles will take the weighted average position of their own historical optimal position $P_i(t)$ and the optimal position of the population $P_g(t)$ as the local attraction point $P_{id}(t)$, and gradually approach this point, the calculation formula is

$$P_{id}(t) = \varphi P_{id}(t) + (1 - \varphi) P_{gd}(t) \quad (15)$$

Where $d = 1, 2, \dots, D$, φ represents the random number in $[0, 1]$.

The position of the particle i in d dimensional space after $t+1$ iterations can be calculated based on the Monte Carlo method as follows:

$$x_{id}(t+1) = P_{id}(t) \pm \beta \times |mbest - x_{id}(t)| \times \ln\left(\frac{1}{u}\right) \quad (16)$$

where β represents the contract-expansion coefficient that is used to adjust the algorithm step size, u represents the random number in $[0, 1]$, and $mbest$ is the mean best position of the population and can be expressed as:

$$mbest = \frac{1}{M} \sum_{i=1}^M P_{id}(t) \quad (17)$$

3.3. Improved quantum particle swarm optimization (IQPSO)

The SOH estimation accuracy using the QPSO will be degraded due to its disadvantages. For example, the initial search range of the population of the QPSO is reduced due to its randomly initialized populations. Moreover, it is easy for the QPSO to fall into local optimum because of its average optimal position ignoring the influence of different particles and the fixed contract-expansion coefficient. In this paper, the IQPSO is developed to estimate the SOH for Li-ion batteries in aspects of the improved population initialization, weighted mean best position, and adaptive contract-expansion coefficient.

3.3.1. Improved population initialization based on the sobol sequence

As one of low-discrepancy sequences, the Sobol sequence can attain a uniform distribution in the probability space and obtain some random numbers to increase the diversity of particle population, which helps to expand the initial search range of the population and improve the global search ability. The initial position of the population f_n can be attained by the Sobol sequence as follows:

$$f_n = f_{\min} + K_n(f_{\max} - f_{\min}) \quad (18)$$

where $f_{\min}$ and $f_{\max}$ are the upper and lower bounds of the hyper-parameters of the LSTM, respectively, and K_n is the random number generated by the Sobol sequence in $[0, 1]$.

3.3.2. Weighted mean best position

In this paper, a quantum particle swarm with weight coefficient set is used to sort according to the fitness value of the particles. One weight coefficient is assigned to each particle. The weight coefficient decreases linearly with the fitness value of the particles from 1.5 to 0.5. The closer the particle to the optimal solution, the greater the weight coefficient assigned. The weighted mean best position is attained as follows:

$$\begin{aligned} mbest(t) &= \frac{1}{M} \sum_{i=1}^M a_{id} P_{id}(t) \\ &= \left(\frac{1}{M} \sum_{i=1}^M a_{i,1} P_{i,1}(t), \frac{1}{M} \sum_{i=1}^M a_{i,2} P_{i,2}(t), \dots, \frac{1}{M} \sum_{i=1}^M a_{i,d} P_{i,d}(t) \right) \end{aligned} \quad (19)$$

3.3.3. Adaptive contraction-expansion coefficient

In the QPSO, the contraction-expansion coefficient β is the only parameter to control the convergence speed of the algorithm. In this paper, an adaptive contract-expansion coefficient is expressed as follows:

$$\beta = \beta_0 + (\beta_1 - \beta_0) \times \left(\frac{t_{\max} - t}{t_{\max}} + \eta \right) \quad (20)$$

where β_0 and β_1 are the maximum and minimum values of β , β_0 and β_1 are set as 0.5 and 1 in this paper, t and $t_{\max}$ are the current and maximum number of iterations, respectively. η is a distance factor, which represents the distance between the current particle and the optimal particle. The η can be expressed as follow

$$\eta = \frac{fit(g)}{fit(i)} \quad (21)$$

where $fit(g)$ and $fit(i)$ are the fitness values of the optimal particle and the current particle, respectively. According to formula (20), β can be dynamically adjusted with the change of particle fitness due to the η , which accelerates the convergence speed of the QPSO.

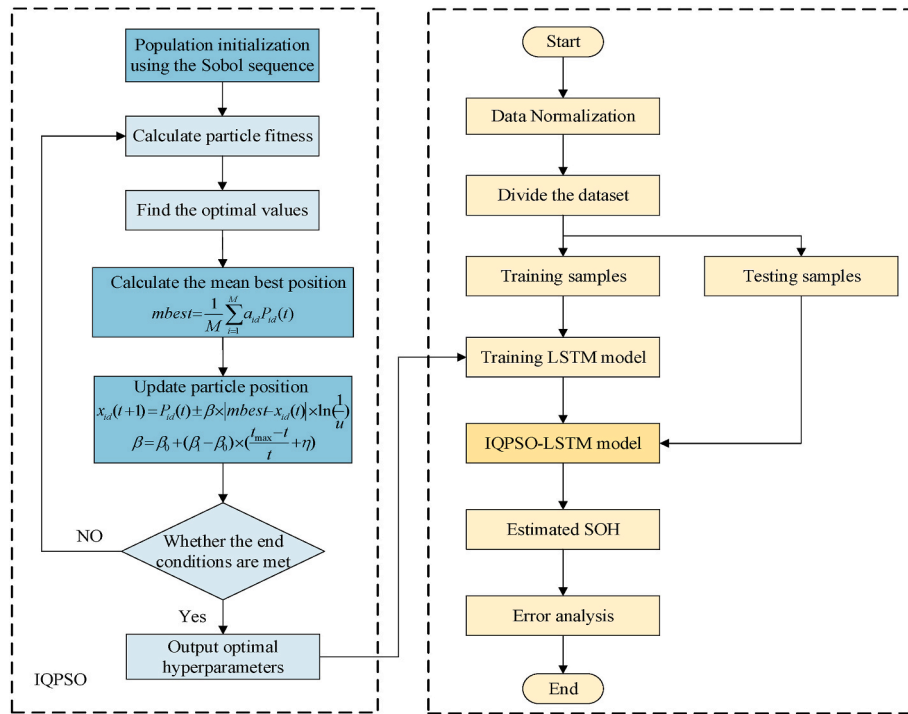


Fig. 7. The flowchart of the developed IQPSO-LSTM.

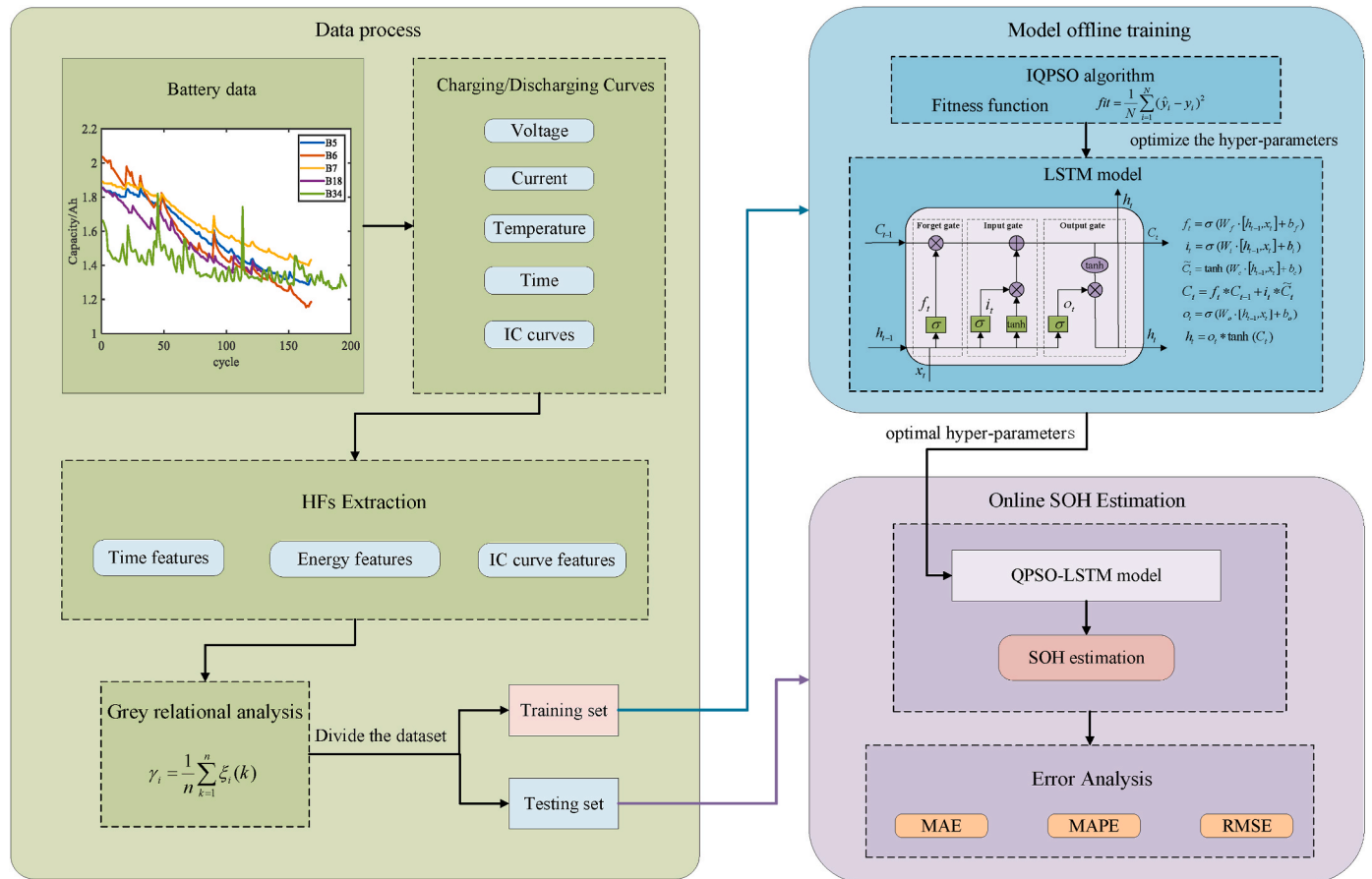


Fig. 8. Framework of the SOH estimation based on IQPSO-LSTM and HFs extraction.

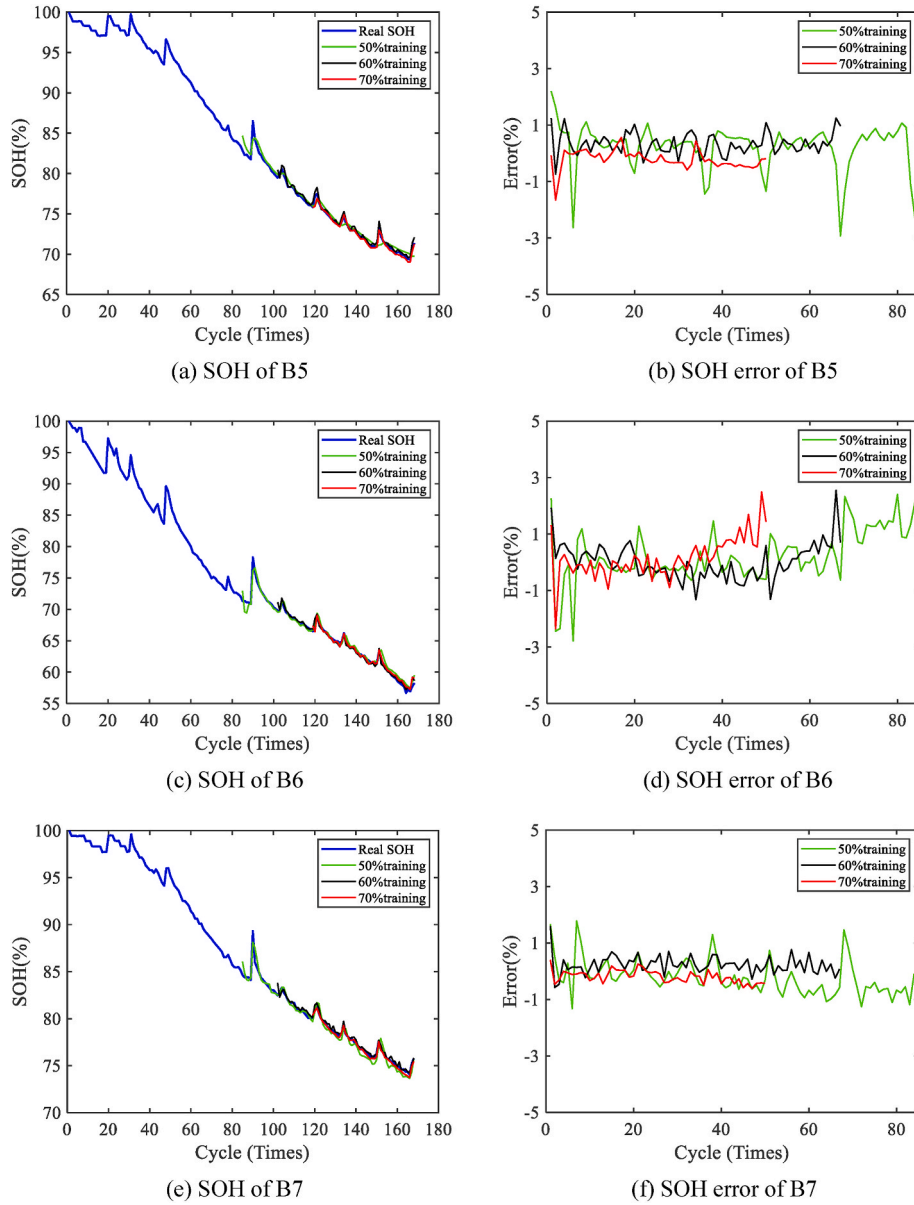


Fig. 9. SOH estimation for different batteries in different training data.

3.4. Developed IQPSO-LSTM

In this paper, the IQPSO is used to optimize the hyper-parameters of the LSTM model, such as the number of hidden layer units, learning rate, iterations, and dropout rate. Compared with the standard QPSO, the IQPSO can overcome the problem of falling into the local optimal solution and improve the global search ability and convergence speed of the QPSO. The flowchart of the IQPSO-LSTM method is shown in Fig. 7.

As shown in Fig. 7, the main steps of the developed IQPSO-LSTM in this paper are presented as follows.

- (1) Normalize the sample data, and allocate the training set and the test set according to the specified proportion, the normalized expression is as follows:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (22)$$

where x and x' are the original sample data and the normalized data,

respectively; $x_{\min}$ and $x_{\max}$ are the minimum and maximum data in the sample data, respectively.

- (2) Initialize the model parameters and the range of LSTM hyper-parameter. For example, the particle population, the maximum number of iterations, and the particle dimension are set as 20, 50, and 4; the number of hidden layer units is [1200], the learning rate is [0.001, 0.01], the number of iterations is [10,200], and the dropout is [0,1].
- (3) Calculate the fitness value of each particle and sort them according to the fitness value, the fitness function is defined as:

$$fit = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2 \quad (23)$$

where N is the number of training samples, $\hat{y}_i$ and y_i are the estimated and actual values of the i -th training sample, respectively.

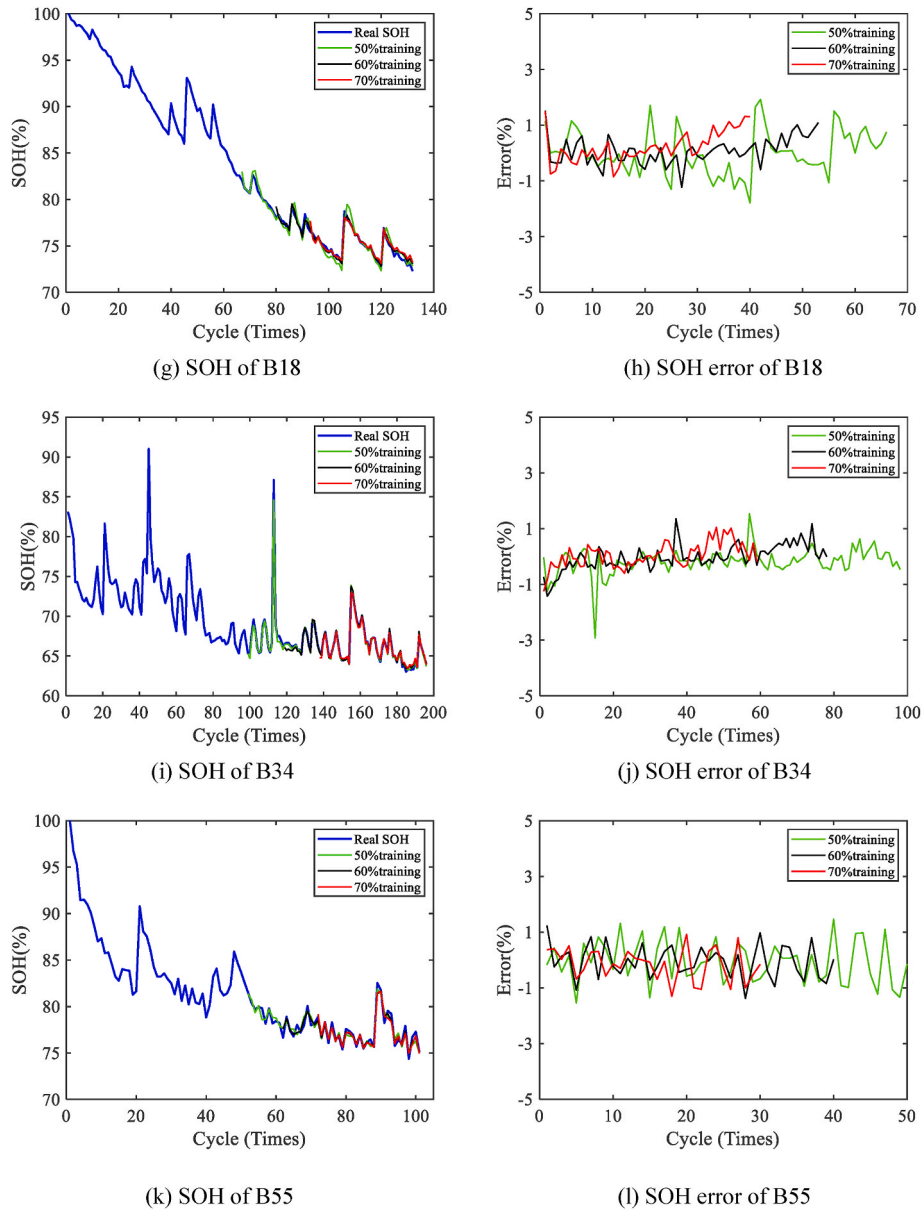


Fig. 9. (continued).

- (4) Attain the individual optimal and group optimal positions of the particles according to the fitness value of each particle, and the optimal position of each particle is used as the historical optimal position.
- (5) Calculate the average optimal position of the particles according to formula (19).
- (6) Update the particle position, the individual optimal and population optimal positions of the particles.
- (7) Determines whether the iteration is completed: if the number of population iterations reaches the maximum number of iterations or the fitness value reaches the minimum value, output the optimal parameter value; otherwise go back to step (4).

3.5. SOH estimation based on IQPSO-LSTM and HF extraction

In this section, a comprehensive SOH estimation framework based on the proposed IQPSO-LSTM method is presented. As is shown in Fig. 8, the proposed SOH framework consists of three parts named data processing, model training, and SOH estimation.

First, the data processing is composed of HFs extraction and correlation analysis, ten HFs including time HFs, energy HFs, and IC HFs are extracted from the charging and discharging curves based on the dataset obtained from the NASA dataset. After that, the correlation between ten HFs and SOH is evaluated by the grey relational analysis. After the above process, the ten HFs are used as the input of the developed IQPSO-LSTM method. The experimental data is divided into training dataset and testing dataset, the training set is used for model training, and the optimal hyper-parameters of the LSTM model are obtained based on the IQPSO during the training process. Finally, the testing data is used for SOH estimation based on the IQPSO-LSTM method.

4. Experimental results and analysis

4.1. The evaluation criteria

In order to evaluate the SOH prediction accuracy of the developed method in this paper, the mean absolute error (MAE), mean absolute percentage error (MAPE), and root mean square error (RMSE) are

Table 2

SOH estimation errors for different batteries in different proportions of training data.

Battery	Training data	MAE (%)	MAPE (%)	RMSE (%)
B5	50%	0.4931	0.6485	0.6454
	60%	0.3056	0.4101	0.3871
	70%	0.2109	0.2909	0.2993
B6	50%	0.4494	0.6926	0.6312
	60%	0.3064	0.4817	0.4041
	70%	0.3153	0.5131	0.4615
B7	50%	0.4190	0.5296	0.5219
	60%	0.2357	0.2990	0.3167
	70%	0.1815	0.2368	0.2211
B18	50%	0.4658	0.6080	0.6085
	60%	0.3018	0.3988	0.3882
	70%	0.3368	0.4518	0.4519
B34	50%	0.2290	0.3332	0.3799
	60%	0.2311	0.3481	0.3064
	70%	0.2342	0.3573	0.2926
B55	50%	0.5124	0.6585	0.6058
	60%	0.3874	0.4985	0.4705
	70%	0.3698	0.4754	0.4654

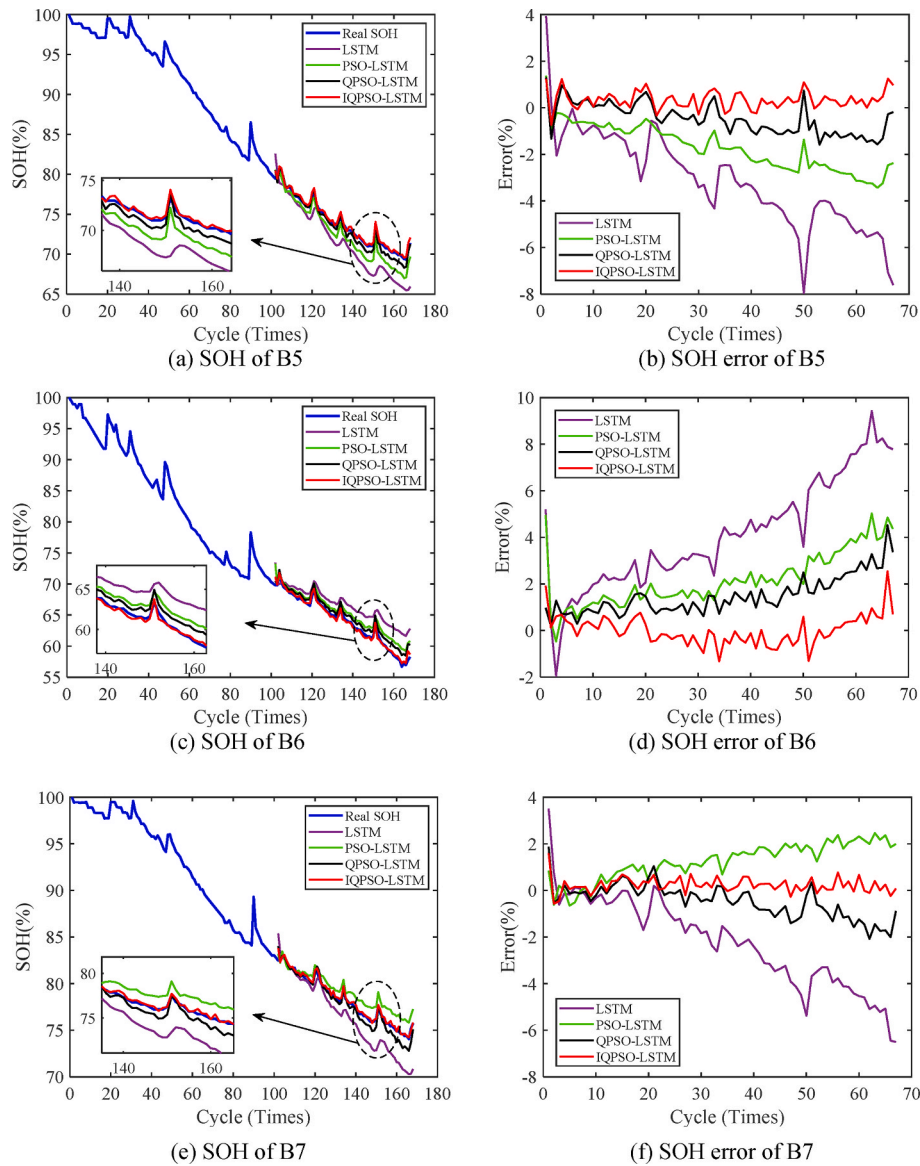
selected, which are expressed as follows:

$$MAE = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i| \quad (24)$$

$$MAPE = \frac{1}{N} \sum_{i=1}^N \frac{|\hat{y}_i - y_i|}{y_i} \quad (25)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2} \quad (26)$$

where N is the number of samples in the testing set, $\hat{y}_i$ and y_i are the estimated and actual SOH values of the i -th training sample, respectively. Smaller values of MAE, MAPE, and RMSE indicate that the predicted value of the model is closer to the true value, and the model estimation is better.

**Fig. 10.** SOH estimation for different batteries using different methods.

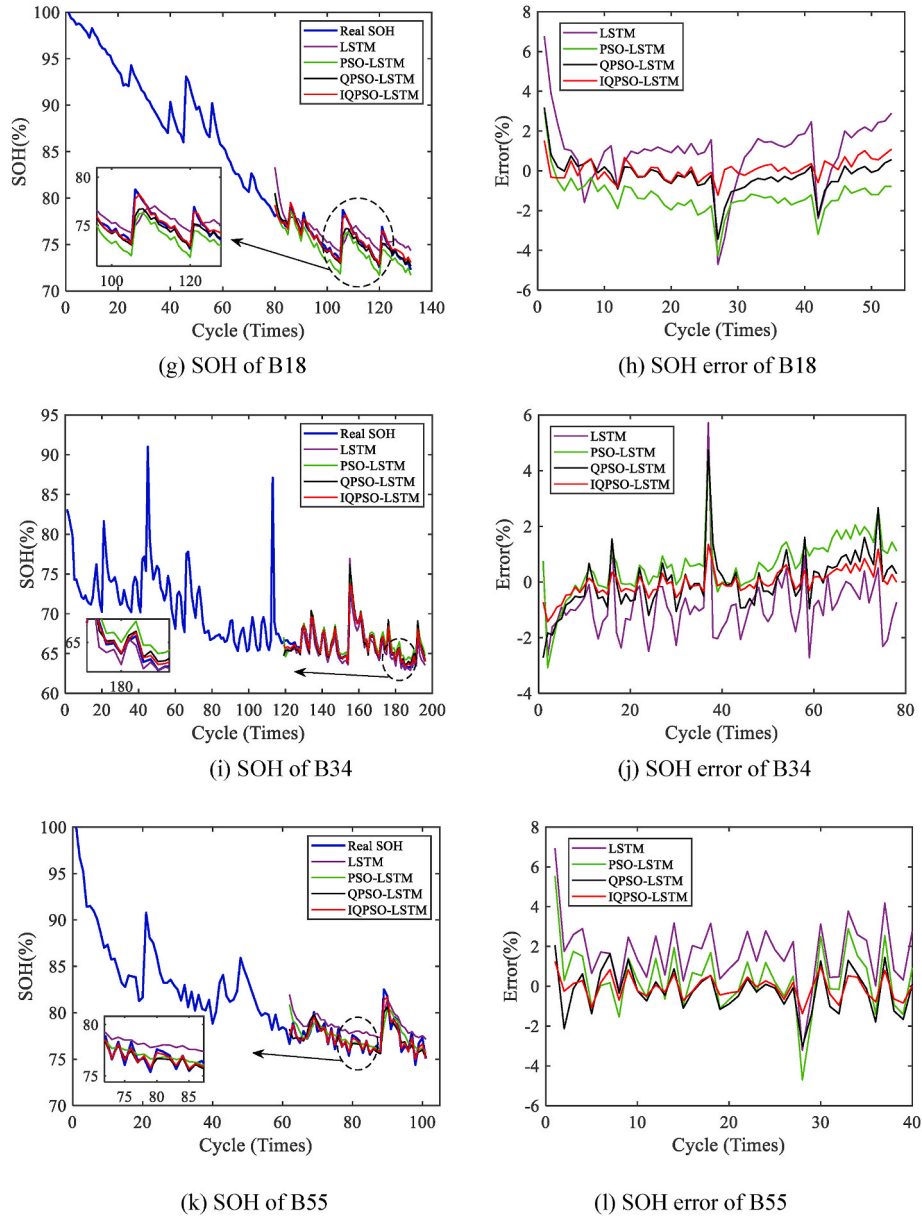


Fig. 10. (continued).

4.2. SOH estimation accuracy based on the developed method using different batteries in different proportions of training data

In this section, different batteries named B5, B6, B7, B18, B34, and B55 in the NASA dataset are used to verify the accuracy using the developed method. Different proportions (50%, 60%, and 70%) of each battery's dataset are selected as the training set, and the rest of the data are chosen as the test set. Fig. 9 shows the SOH estimation results and its corresponding errors for different batteries using 50%, 60%, and 70% training set data, respectively. It can be seen that the estimated SOH using the developed method with different proportions of training data can all exactly follow the real SOH; the capacity regeneration phenomenon during the ageing process can be captured as well. For example, as shown in Fig. 9 (i), in spite of the capacity fluctuation amplitude of B34 being the largest one among the other batteries in 4 A discharging current, the estimated SOH using the developed method can accurately match the real SOH. It means that the developed method can accurately predict the SOH of the batteries under different discharging conditions. In addition, Fig. 9 (k) shows the estimated results of B55 at a

low temperature of 4 °C. It can be seen that the developed method can accurately predict the SOH of the batteries under different temperature conditions with high robustness and generalization. Moreover, as shown in Fig. 9, the relative errors of the estimated SOH for each battery are all kept within a stable range between −1% and 1%. The maximum relative error does not exceed 3%. It is further proved that the SOH can be estimated accurately using the developed method with high accuracy for different batteries in spite of the proportions of training data.

Table 2 shows the SOH estimation errors using the developed method for different batteries in different proportions of training data. It can be seen that all the errors are less than 1%, and most of the errors are within 0.5%. Moreover, the maximum values of the MAE, the MAPE, and the RMSE are 0.5124%, 0.6926%, and 0.6454%, respectively. Besides, the maximum errors in different batteries are always attained in 50% of the training data. This is because the fewer training data there are, the less information about the battery degradation history is available, resulting in a decrease in the accuracy of the constructed IQPSO-LSTM method. It indicates that the later SOH estimation can show higher accuracy compared to the early SOH estimation in battery

Table 3
SOH estimation errors for different batteries using different methods.

Battery	Method	MAE (%)	MAPE (%)	RMSE (%)
B5	IQPSO-LSTM	0.3056	0.4101	0.3871
	QPSO-LSTM	0.5055	0.6945	0.5961
	PSO-LSTM	1.2645	1.7453	1.404
	LSTM	2.3507	3.241	2.6819
B6	IQPSO-LSTM	0.3064	0.4817	0.4041
	QPSO-LSTM	0.9216	1.4768	1.0237
	PSO-LSTM	1.3688	2.1926	1.5168
	LSTM	2.5907	4.1675	2.8702
B7	IQPSO-LSTM	0.2357	0.2990	0.3167
	QPSO-LSTM	0.5037	0.6549	0.6508
	PSO-LSTM	1.0152	1.3168	1.1262
	LSTM	1.8101	2.362	2.2456
B18	IQPSO-LSTM	0.3018	0.3988	0.3882
	QPSO-LSTM	0.4280	0.5589	0.6888
	PSO-LSTM	1.0673	1.4103	1.1981
	LSTM	1.1934	1.5772	1.4865
B34	IQPSO-LSTM	0.2311	0.3481	0.3064
	QPSO-LSTM	0.4876	0.7299	0.6961
	PSO-LSTM	0.5667	0.8574	0.7859
	LSTM	0.7654	1.1396	0.9689
B55	IQPSO-LSTM	0.3874	0.4985	0.4705
	QPSO-LSTM	0.6991	0.8961	0.8828
	PSO-LSTM	0.9621	1.2411	1.309
	LSTM	1.1451	1.8881	1.7886

health management, which can provide the guidance on the time to replace the batteries.

4.3. Compared with other methods

In order to further verify the effectiveness and superiority of the developed method, the comparisons of the SOH estimation accuracy using the LSTM, the PSO-LSTM, the QPSO-LSTM, and the developed method are carried out in different batteries. In these tests, 60% of each battery's data is used for the model training, and 40% of each battery's data is selected for the model testing. The number of the particle populations for the PSO, the QPSO, and the IQPSO is set as 20, and the number of iterations and the particle dimension are set as 50 and 4, respectively. The hyper-parameters of the single LSTM model are set as follows: the number of hidden layer units is 100, the learning rate is 0.005, the number of iterations is 100, and the dropout rate is 0.5.

The SOH estimation results and relative errors of different batteries using the four methods are shown in Fig. 10. It can be seen that the estimated SOH using the IQPSO-LSTM can always capture the actual SOH. Moreover, compared with the other three methods, the relative error of the SOH estimation based on the developed method is minimum, especially in capacity regeneration and local fluctuations process.

Table 3 lists the MAE, MAPE, and RMSE of the SOH estimation for different batteries using different methods. It can be seen that the SOH can be accurately estimated using the developed method for the different batteries with the minimal value in terms of the MAE, the MAPE, and the RMSE. Taking the B5 battery as an example, the MAE, the MAPE, and the RMSE of the developed method are 0.3056%, 0.4101%, and 0.3871%, respectively; the MAE, the MAPE, and the RMSE of the QPSO-LSTM are 0.5055%, 0.6945%, and 0.5961%, respectively; the MAE, the MAPE, and the RMSE of the PSO-LSTM are 1.2645%, 1.7453%, and 1.404%, respectively; and the MAE, the MAPE, and the RMSE of the LSTM are 2.3507%, 3.241%, and 2.6819%, respectively. The SOH estimation accuracy based on the LSTM is degraded because the hyper-parameters of the LSTM, such as the learning rate, are constant and cannot be adjusted dynamically. The PSO and the QPSO can continuously adjust the hyper-parameter of the LSTM according to the model error. However, the accuracy of the model is limited because the above two algorithms will fall into local optimal and cannot output optimal hyper-parameters. However, the SOH estimation using the developed method can avoid the local optimal solution and output the optimal

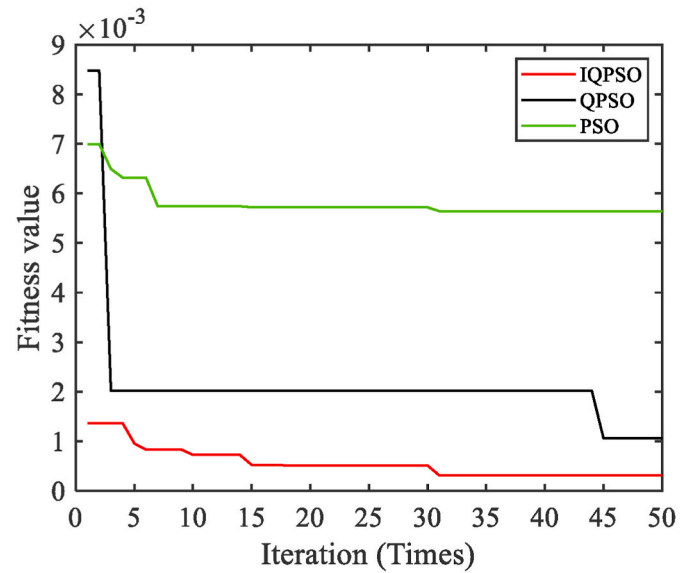


Fig. 11. The optimization process of three algorithms.

LSTM hyper-parameters due to the improved strategies of population initialization using the Sobol sequence, weighted mean best position, and adaptive contract-expansion coefficient. Taking the B5 battery as an example, the optimization process and hyper-parameter optimization results using the three algorithms are shown in Fig. 11. It can be seen from Fig. 11 that IQPSO has the lowest fitness value (below 0.001) in the stable phase, which indicates that it has the best optimization effect.

In general, the evaluation indexes of the six batteries obtained by different methods are kept at a low level, with the maximum MAE, MAPE, and RMSE less than 5%. It proves that the HFs selected in this article can well reflect the degradation of the different batteries and can be used in the SOH estimation methods.

5. Conclusions

The SOH will be inaccurately estimated using data-driven methods when a single HF is extracted with incorrect hyper-parameters selection. A SOH estimation method based on multi-health features extraction and IQPSO-LSTM is developed for Li-ion batteries in this paper. Firstly, ten HFs divided into three categories—time HFs, energy HFs, and IC HFs—are extracted from the batteries' data to effectively and comprehensively describe the batteries' degradation. Secondly, the correlations between the HFs and the SOH are evaluated by the grey correlation analysis. To avoid falling into local optimization in the QPSO and PSO and output the optimal hyper-parameters, this paper presents the IQPSO based on the improved strategies of population initialization using the Sobol sequence, weighted mean best position, and adaptive contract-expansion coefficient.

The accuracy and effectiveness of the SOH estimation based on the developed method is verified by the different batteries with different discharge conditions and different temperature conditions using the NASA battery dataset. The results showed that the SOH of the Li-ion batteries can be estimated by the developed method with high accuracy and robustness under different training data. In different batteries named B5, B6, B7, B18, B34, and B55 in different proportions of training data (50%, 60%, and 70%), the maximum relative error of the SOH estimation using the developed method is within 3%, and the corresponding MAE, MAPE, and RMSE of the SOH are all less than 1%. Moreover, compared with other methods such as the OPSO-LSTM, PSO-LSTM, and LSTM, the SOH can be accurately estimated using the developed method with the minimal value in terms of the MAE, MAPE, and RMSE. Moreover, the maximum MAE, MAPE, and RMSE of the SOH

based on the developed method are less than 0.5%.

In the future, some experiments in different types of batteries will be conducted to verify the effectiveness of the developed method. In addition, the future research will focus on battery packs and estimate the SOH of battery packs based on battery data from practical application scenarios, to meet the actual demands in the fields of energy storage and electric vehicles.

Credit author statement

Simin Peng: Conceptualization, Supervision, Writing – original draft. **Yunxiang Sun:** Formal analysis, Investigation, Methodology. Writing – original draft. **Dandan Liu:** Software, Data curation, Visualization. **Quanqing Yu:** Conceptualization, Supervision, Writing – Reviewing and Editing. **Jiarong Kan:** Software and Data curation. **Michael G Pecht:** Visualization, Writing – Reviewing and Editing.

Declaration of competing interest

We confirm that neither the manuscript nor any parts of its content are currently under consideration or published in another journal. All authors have approved the manuscript and agree with its submission to Energy. Both authors are fluent English speakers, with considerable expertise in the field of BESS control and battery management system. We have no conflicts of interest to declare. Do not hesitate to contact us if you have any questions about the manuscript.

Data availability

Data will be made available on request.

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